Productivity and Growth

Reference: Macroeconomics in Modules
by Krugman and Wells

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Sustained Economic Growth

- A *sustained increase* in output per capita.
- Sustained long term growth depends on rising productivity.
- Productivity (Labor productivity refers to output per worker).
- Needed for GDP growth.

Aggregate Production Function

$$Y = f(L, K)$$

More specifically,

$$Y = aL^{\alpha}K^{\beta}$$

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- a is a positive constant that reflects technology.
- L is labour; K is capital.
- α is the share or elasticity of output w.r.t. labor.
- β is the share or elasticity of output w.r.t. capital.
- $\alpha + \beta =$?

$Y = aL^{\alpha}K^{\beta}$ in per capita terms

- What would you do? If $\alpha + \beta = 1$, what is the clever way to do it?

$Y = aL^{\alpha}K^{\beta}$ in per capita terms

- We do this to get $y = ak^{\beta}$, where the small letters represent per capita variables.
- We focus primarily on y and k.

Explaining growth in Productivity

- Increase in physical capital per worker $\left(\frac{K}{L}\right)$.
- Increase in human capital per worker $\left(\frac{H}{L}\right)$.
- Technological progress (A).

Natural Resources

- Not as important as productivity.
- Resource Curse.
  - Dutch disease.
  - Immiserizing growth.

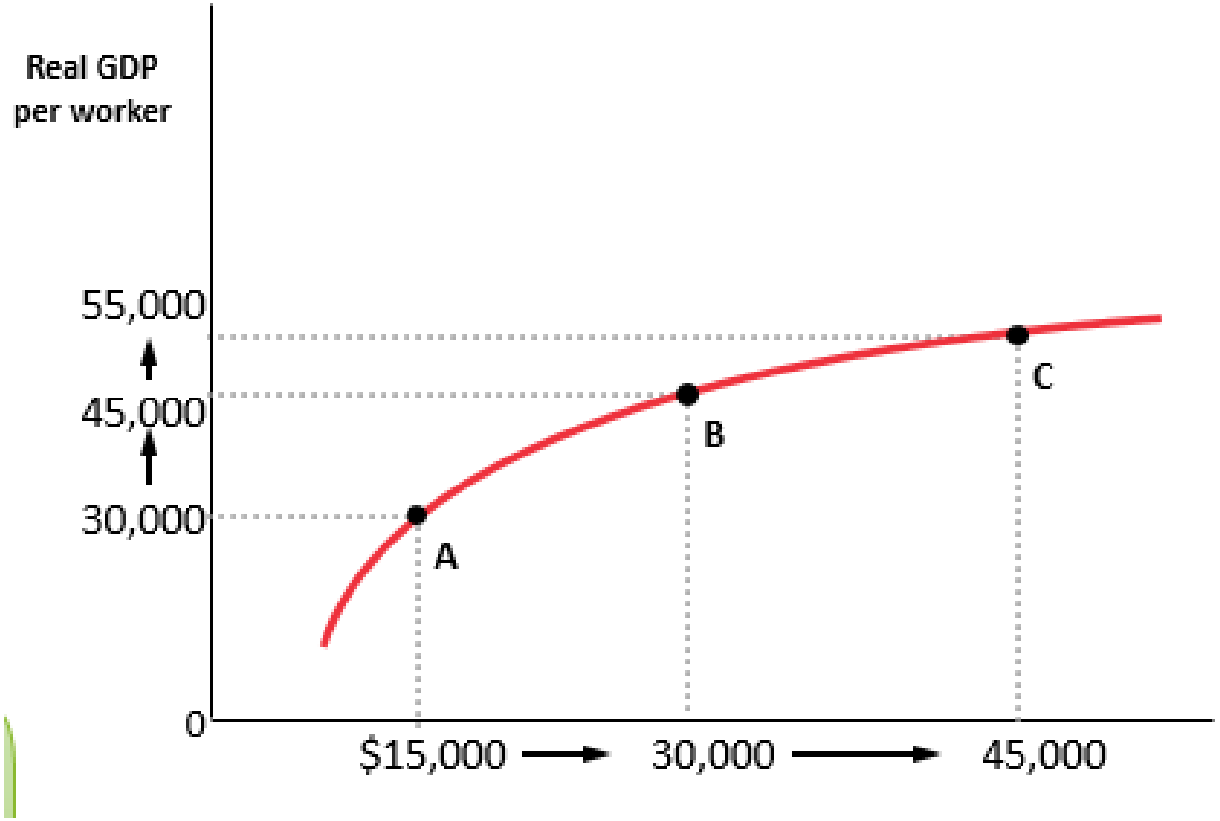
Diminishing Returns

- *Diminishing returns to physical capital:* Holding the amount of human capital per worker and the state of technology fixed, each successive increase in the amount of physical capital per worker leads to a smaller increase in productivity.

Physical capital per worker	Real GDP per worker
$0	$0
15,000	30,000
30,000	45,000
45,000	55,000

**Source: Macroeconomics in Modules by Paul Krugman and Robin Wells

Diminishing Returns to Physical Capital



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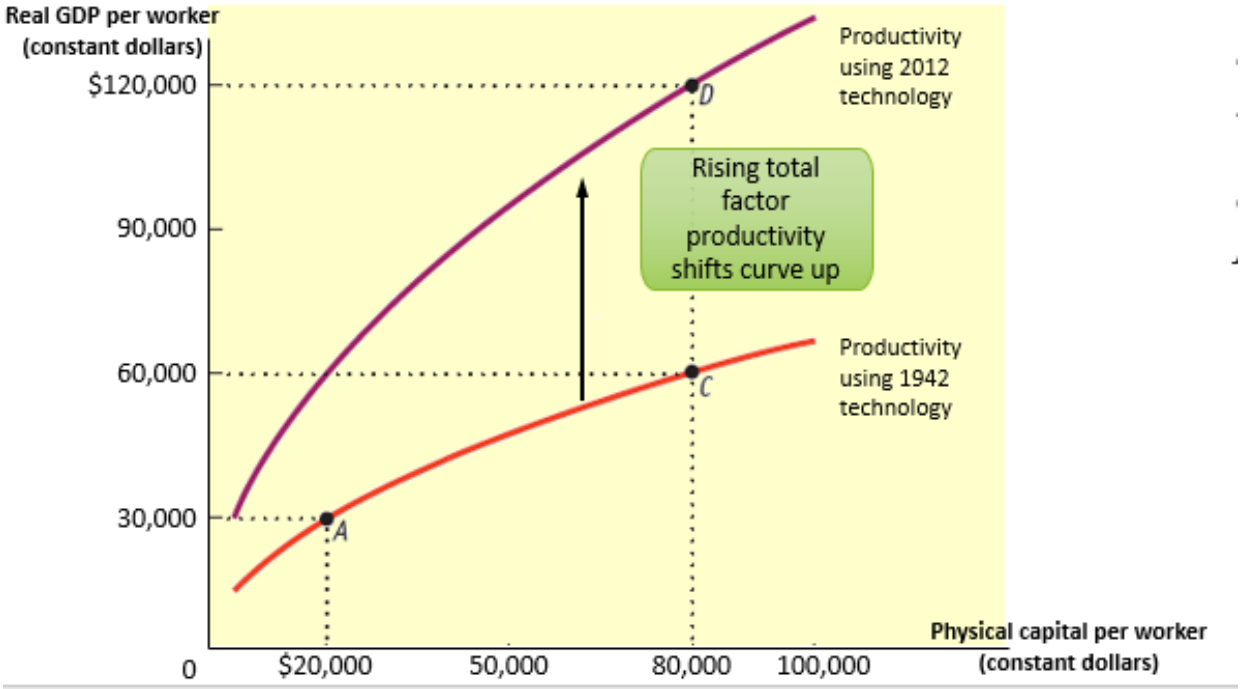
Total Factor Productivity (TFP)

- *Total factor productivity* is the amount of output that can be achieved with a given amount of factor inputs.



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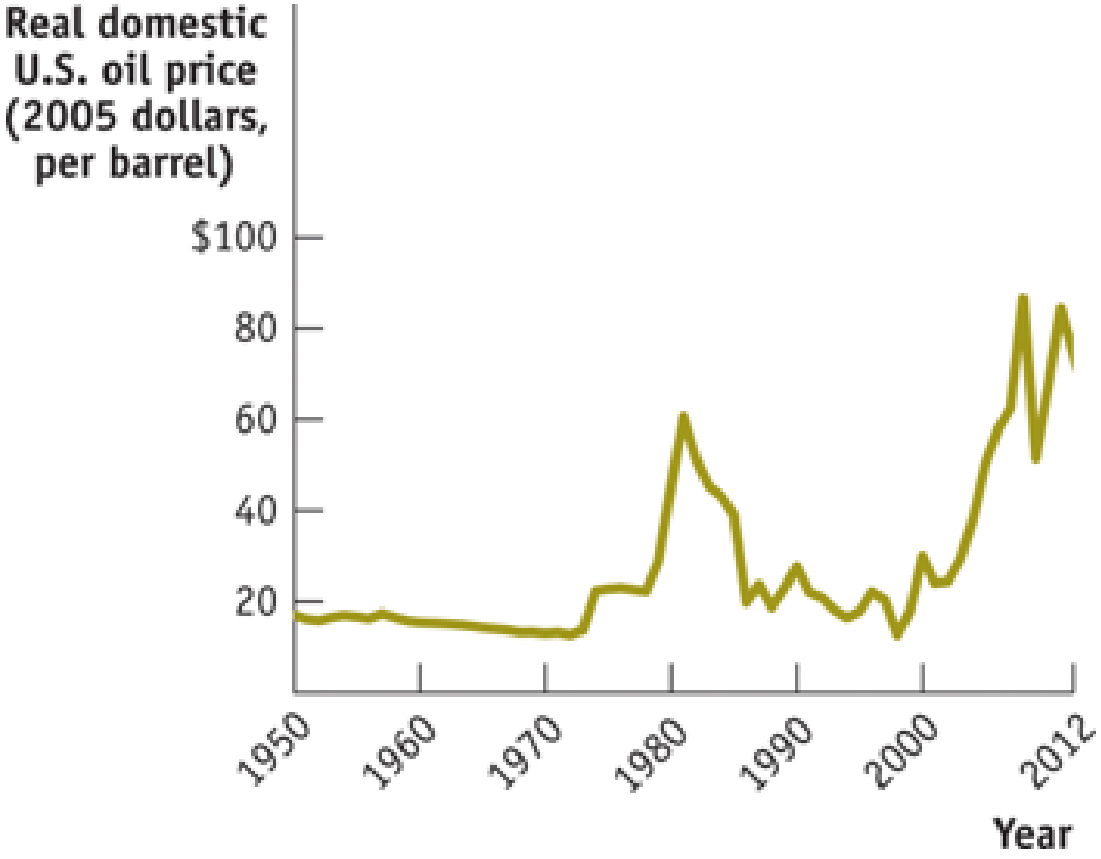
Is World Growth Sustainable?

- *Sustainable long-run economic growth:* Long-run growth that can continue in the face of the limited supply of natural resources and the impact of growth on the environment.
- What is the impact of limited natural resources on long-run economic growth? Three important questions:
  - How large are the supplies of key natural resources?
  - How effective will technology be at finding alternatives to natural resources?
  - Can long-run economic growth continue in the face of resource scarcity?

Real Prices of Oil, 1949-2012



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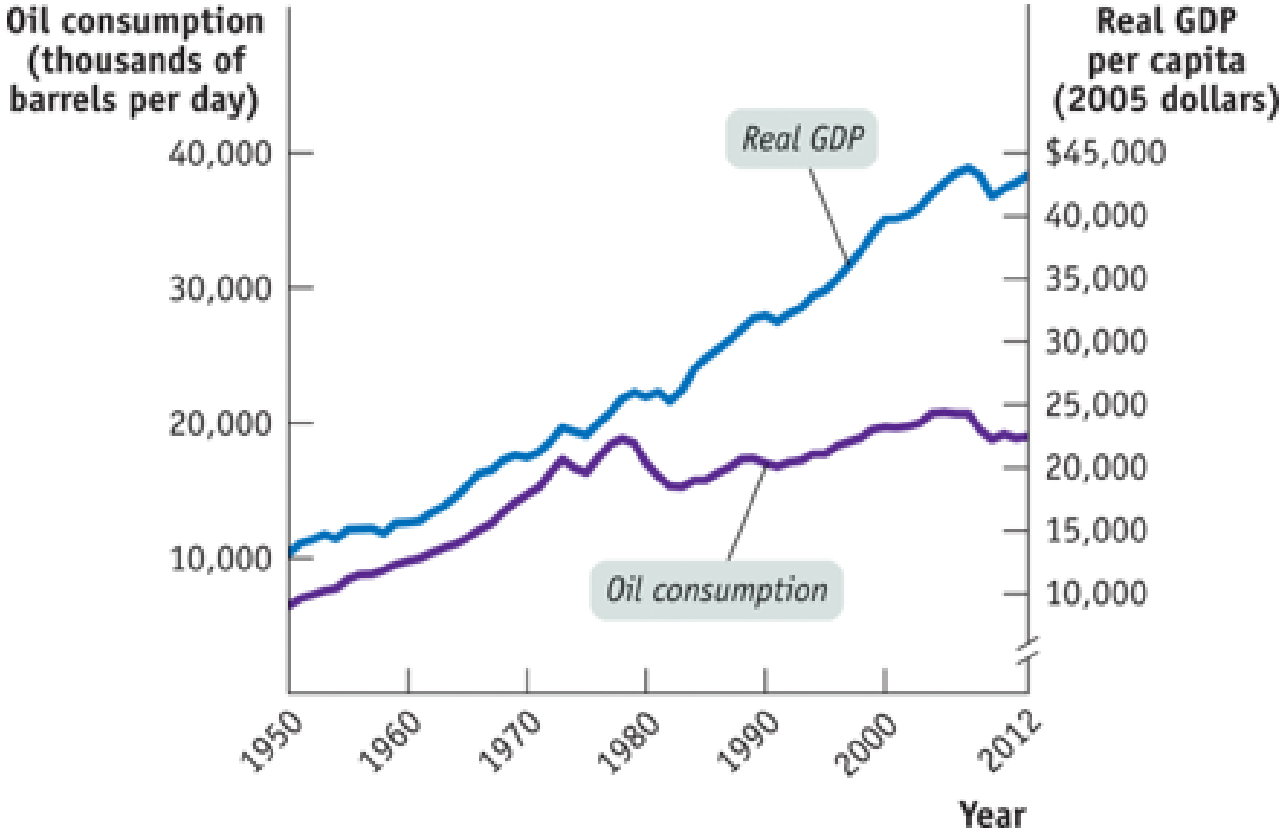
U.S. Oil Consumption and Growth over Time



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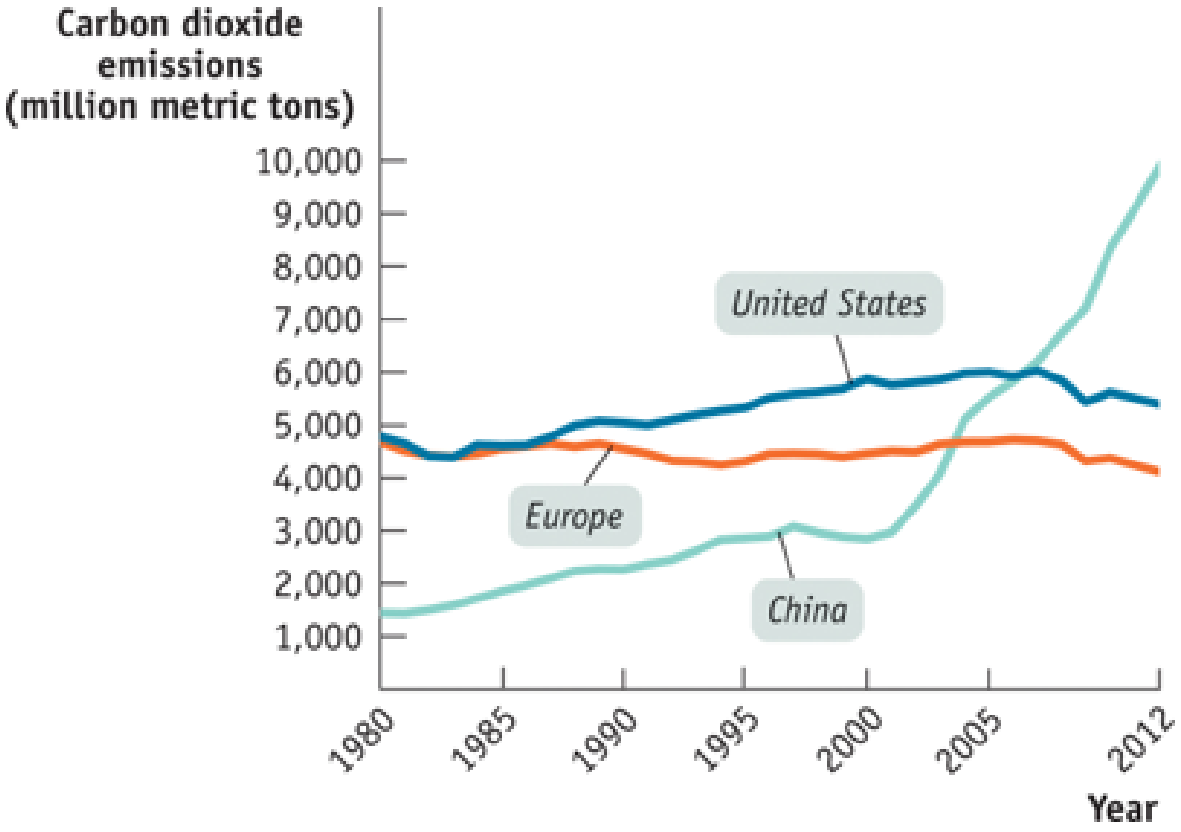
Have a look here for more details.

Climate Change and Growth



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